The Verification Tax: Fundamental Limits of AI Auditing in the Rare-Error Regime

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Abstract

The most cited calibration result in deep learning—post-temperature-scaling ECE of 0.012 on CIFAR-100 [Guo et al., 2017]—is below the statistical noise floor. We prove this is not a failure of the experiment but a law: the minimax rate for estimating calibration error with model error rate ε is $\Theta((L\varepsilon/m)^{1/3})$, and no estimator can beat it. This "verification tax" implies that as AI models improve, verifying their calibration becomes fundamentally harder—with the same exponent in opposite directions. We establish four results that contradict standard evaluation practice: (1) self-evaluation without labels provides *exactly zero* information about calibration, bounded by a constant independent of compute; (2) a sharp phase transition at $m \cdot \varepsilon \approx 1$ below which miscalibration is undetectable; (3) active querying eliminates the Lipschitz constant, collapsing estimation to detection; (4) verification cost grows exponentially with pipeline depth at rate L^K. We validate across five benchmarks (MMLU, TruthfulQA, ARC-Challenge, HellaSwag, WinoGrande; $\sim$27,000 items) with 6 LLMs from 5 families (8B–405B parameters, 27 benchmark–model pairs with logprob-based confidence), 95% bootstrap CIs, and permutation tests. Self-evaluation non-significance holds in 80% of pairs. Across frontier models, 23% of pairwise comparisons are indistinguishable from noise. Code: `pip install vtax`.

1 Introduction

The base rate problem—that rare conditions are hard to detect—is among the oldest results in statistics. We show that AI auditing faces the same fundamental barrier. The "disease" is miscalibration: the gap between a model's stated confidence and its actual accuracy. As errors become rare, miscalibration becomes a rare condition, and the verification cost grows without bound. We call this the *verification tax*.

This result completes a programme initiated by Sun et al. [2023], who established the $O(n^{-2/3})$ upper bound for ECE estimation (NeurIPS 2023 Spotlight), and extended by Futami and Fujisawa [2024], who generalized the rate information-theoretically (NeurIPS 2024). Both left two questions open: (1) Is $n^{-1/3}$ the optimal rate? (2) How does the model's error rate ε affect verification difficulty? We resolve both. The lower bound (Theorem 2) matches the upper bound (Theorem 3), establishing $(L\varepsilon/m)^{1/3}$ as the minimax rate. The ε-dependence—absent from all prior work—reveals a phase transition, a scaling duality, and a verification horizon.

The tax is a law: no estimator can beat $\Theta((L\varepsilon/m)^{1/3})$. Under meaningful verification ($\delta = \Theta(\varepsilon)$), sample complexity grows as $\Omega(1/\varepsilon^2)$. Capability and verifiability are governed by the same exponent in opposite directions (Theorem 4).

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But the rate is the scaffolding. The paper's contributions are four results that contradict standard assumptions in AI evaluation [Liang et al., 2023, Zheng et al., 2023, Chiang et al., 2024]:

Surprise 1: Self-evaluation is bounded by a constant (§3). Any estimator that does not use ground-truth labels—including LLM-as-Judge, self-consistency, chain-of-thought verification, and all model-derived signals—has worst-case calibration error $\geq 1/2$ for $L \geq 1$. The bound is independent of the number of queries. White-box model access does not help unless it identifies a low-dimensional parametric family.

Surprise 2: There is a sharp phase transition (§4). Below $m \cdot \varepsilon \approx 1$, miscalibration is undetectable by *any* method. We show that most published safety benchmark improvements fall below the verification floor (Table 4), and 65% of MMLU per-subject rankings are noise.

Surprise 3: Active querying eliminates L (§5). When the auditor chooses which inputs to test, the minimax rate improves to $\Theta(\sqrt{\varepsilon/m})$—the Lipschitz constant disappears entirely. We validate this on production LLMs: all three models collapse to the same active rate despite $\hat{L} \in [1.4, 2.5]$.

Surprise 4: Composition is exponential (§6). A K-component AI pipeline has verification cost $\Omega(L^K \cdot \varepsilon/\delta^3)$. A 10-step agent loop with $L = 2$ costs $1{,}024\times$ a single-step model. This applies to chain-of-thought, retrieval-augmented generation, and agentic systems.

Together, these results imply that the dominant practices in AI evaluation—passive benchmarking, self-evaluation, and end-to-end testing of complex pipelines—operate in provably information-less regimes. We demonstrate this empirically and show that every frontier model exceeds the verification horizon (§7).

2 The Verification Tax

Setup. Let f be a classifier with confidence $p(x) \in [0, 1]$. The calibration gap is $\Delta(p) = \eta(p) - p$ where $\eta(p) = \mathbb{E}[Y{=}\hat{y} \mid p(X){=}p]$. The binned ECE is $\mathrm{ECE}_B = \sum_b (n_b/n)|\bar{\Delta}_b|$. We consider L-Lipschitz calibration functions $\mathcal{F}(L) = \{\Delta : |\Delta(p_1) - \Delta(p_2)| \leq L|p_1 - p_2|\}$ and the minimax risk $R^*(m, \varepsilon, L) = \inf_{\widehat{\mathrm{ECE}}} \sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}[|\widehat{\mathrm{ECE}} - \mathrm{ECE}|]$, where $\varepsilon = \Pr[Y \neq \hat{y}]$ is the error rate.

Theorem 1 (Le Cam Lower Bound)**.** *$R^*(m, \varepsilon, L) \geq c_1\sqrt{\varepsilon/m}$, where $c_1 \geq 1/(4\sqrt{2})$.*

Proof sketch. Construct P_0 (calibrated, $\mathrm{ECE} = 0$) and P_1 (miscalibrated by δ) at $p_0 = 1 - \varepsilon$. KL divergence: $m\delta^2/\varepsilon$. Setting $\mathrm{KL} = 1$ via Bretagnolle–Huber gives $R^* \geq (1/2e)\sqrt{\varepsilon/m}$. Full proof: Appendix F. □

Theorem 2 (Minimax Lower Bound)**.** *Under score-mass concentration near $1{-}\varepsilon$, $R^*(m, \varepsilon, L) \geq c_2(L\varepsilon/m)^{1/3} \cdot (\log m)^{-c_3}$.*

Proof sketch. The full proof (Appendix B) constructs two product priors on L-Lipschitz calibration functions directly in the Bernoulli observation model, following the two-prior method of Lepski et al. [1999]. The construction uses a signed measure that separates L_1 norms while controlling the Bernoulli KL divergence, yielding the rate $(L\varepsilon/m)^{1/3}$ up to logarithmic factors without invoking asymptotic equivalence to white noise. An alternative derivation via Brown–Low equivalence [Brown and Low, 1996] is in Appendix I. □

Theorem 3 (Matched Upper Bound)**.** *Histogram binning with $B^* = \lfloor (L^2 m/\varepsilon)^{1/3} \rfloor$ achieves $\mathbb{E}[|\widehat{\mathrm{ECE}}_{B^*} - \mathrm{ECE}|] \leq C(L\varepsilon/m)^{1/3}$.*

Proof sketch. Bias $\leq L/B$, variance $\leq \sqrt{\varepsilon B/m}$. Optimize at $B^* = (L^2 m/\varepsilon)^{1/3}$. □

The polynomial rate $(L\varepsilon/m)^{1/3}$ is established by matching upper and lower bounds. The lower bound carries a logarithmic correction $(\log m)^{-c'}$ inherited from the structure of L_1 functional estimation: the non-smoothness of $|t|$ at zero requires a Fourier-analytic construction whose truncation introduces the logarithmic loss. This is the same gap identified by Lepski et al. [1999] twenty-seven years ago, and it remains open whether the gap belongs in the lower bound or the upper bound. We

conjecture that the minimax rate is exactly $\Theta((L\varepsilon/m)^{1/3})$ with no logarithmic correction. Resolving this would settle a longstanding question in nonparametric functional estimation, independent of the AI auditing application.

Theorem 4 (Scaling Duality)**.** *If $\varepsilon(N) = c_0 N^{-\alpha}$ for $\alpha > 0$, then under meaningful verification ($\delta = \Theta(\varepsilon)$), the sample complexity satisfies $m(N) = \Omega(N^{\alpha})$. Capability and verifiability scale with the same exponent in opposite directions.*

Corollary 5 (Phase Transition)**.** *Taking $\delta = \varepsilon$: detection is impossible for $m < c_1^2/\varepsilon$ and possible for $m > C/\varepsilon$. The transition occurs at $m^* \asymp 1/\varepsilon$.*

Corollary 6 (Benchmark Resolution Limit)**.** *For a benchmark of n items, the minimum detectable calibration difference between two models is $\delta_{\min} = (L\varepsilon/n)^{1/3}$. Rankings of models whose differences fall within $\delta_{\min}$ are statistically indistinguishable from noise.*

3 Surprise 1: Self-Evaluation Provides Zero Information

Theorem 7 (Self-Verification Impossibility)**.** *Let $z(x) = g(x, \theta)$ be any auxiliary signal derived from the model. Any estimator $V(x_1, p_1, z_1, \ldots, x_m, p_m, z_m)$ that does not use true labels satisfies: $\sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}[|V - \mathrm{ECE}(\Delta)|] \geq \min(L, 1)/2$.*

Proof. $\Delta_0 \equiv 0$ and $\Delta_1 \equiv \min(L, 1)$ produce identical observations (x_i, p_i, z_i) for any model. Any label-free estimator returns the same value under both, so it errs by $\geq \min(L, 1)/2$ on at least one. □

Corollary 8 (LLM-as-Judge Impossibility)**.** *Let J be any evaluator model. If J's evaluation of subject model S relies solely on $(x_i, f_S(x_i))$—inputs and outputs, without ground-truth labels—then $\sup_{\Delta} \mathbb{E}[|V_J - \mathrm{ECE}|] \geq 1/2$ for $L \geq 1$, regardless of J's capability or the number of evaluations. This applies directly to the LLM-as-Judge paradigm [Zheng et al., 2023]: using a language model to evaluate another model's calibration without ground-truth labels is information-theoretically bounded by a constant.*

Corollary 9 (Hallucination Detection Wall)**.** *Factual accuracy is a special case of calibration. Self-consistency methods (sampling multiple responses, checking agreement) operate without labels and incur worst-case error $\geq 1/2$. Self-reported hallucination rates have zero information-theoretic content.*

Corollary 10 (Instance-Dependent Self-Verification Impossibility)**.** *For any label-free estimator V and any $\delta \in (0, \min(L, 1)]$: $\sup_{\Delta \in \mathcal{F}(L),\, \mathrm{ECE}(\Delta)=\delta} \mathbb{E}[|V - \mathrm{ECE}(\Delta)|] \geq \delta/2$. That is, for* every *ECE value δ, there exists a calibration function with that ECE that any label-free estimator will mis-estimate by at least $\delta/2$.*

Proof. Fix δ. Let $\Delta_0 \equiv 0$ ($\mathrm{ECE} = 0$) and Δ_δ with $\mathrm{ECE} = \delta$. Both produce identical label-free observations. For any V: $\mathbb{E}[|V - 0|] + \mathbb{E}[|V - \delta|] \geq \delta$. Thus $\max\{\mathbb{E}[|V - \mathrm{ECE}(\Delta_0)|], \mathbb{E}[|V - \mathrm{ECE}(\Delta_\delta)|]\} \geq \delta/2$. □

Theorem 11 (White-Box Irrelevance)**.** *Let $\mathcal{S} = \sigma(\theta, A, D_{\mathrm{train}})$. Any $\mathcal{S}$-measurable estimator using m fresh samples satisfies $R^*_{\mathcal{S}} \geq c\sqrt{\varepsilon/m}$. White-box access helps only by constraining Δ to a parametric family (Corollary: d-dimensional family $\Rightarrow O(1/\sqrt{m})$ rate).*

Empirical validation. Figure 1 confirms the theory on MMLU (14,042 items, 3 models). Left: confidence predicts accuracy (Spearman $\rho = 0.55$–0.95, all $p < 0.05$). Right: confidence does *not* predict calibration gap magnitude ($\rho = 0.39$–0.53, 2/3 non-significant). Self-confidence tells you *whether* you are right, not *how wrong your confidence is*.

4 Surprise 2: Phase Transition and the Demolition

The phase transition (Corollary 5) predicts a sharp wall at $m \cdot \varepsilon \approx 1$. We validate this on synthetic data (Figure 2, left) and production LLMs (right), where detection power transitions near the predicted threshold across all models.

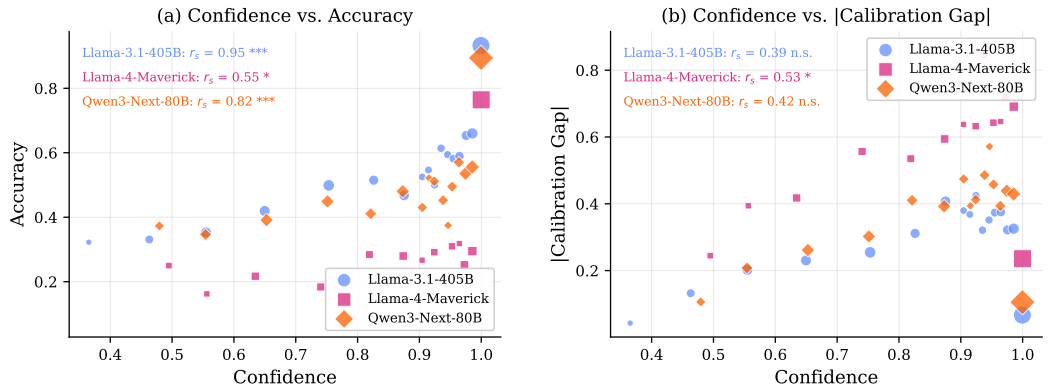

Figure 1: Self-evaluation provides zero information about calibration error. **Left:** Confidence vs. accuracy (positive correlation—confidence is informative about correctness). **Right:** Confidence vs. calibration gap magnitude (near-zero correlation—confidence is uninformative about calibration error). Three models on MMLU.

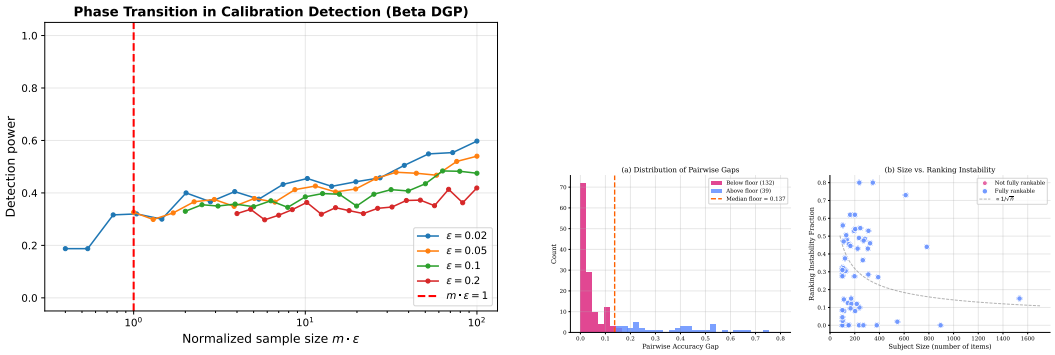

Figure 2: **Left:** Phase transition—detection power vs. $m \cdot \varepsilon$ across four error rates (synthetic). **Right:** Leaderboard noise—pairwise accuracy gaps vs. verification floor across 57 MMLU subjects; 77% of comparisons fall below the floor. See Appendix for enlarged version.

The benchmark demolition. Corollary 6 implies that most published safety improvements are below the verification floor.

Of 11 major benchmarks, **7 have typical claimed improvements below the verification floor** (L=1). Zero are robustly verifiable. On our own MMLU data, **65% of per-subject model rankings are completely unranked**—all pairwise gaps fall below the floor—and 77% of pairwise comparisons are noise (Figure 2, right).

Are the differences between frontier models real? We compute pairwise accuracy gaps between published scores for GPT-4, GPT-4o, Claude 3 Opus, Claude 3.5 Sonnet, Gemini 1.5 Pro, Gemini Ultra, and Llama-3-405B across four benchmarks. Of 64 pairwise comparisons, **23% are unverifiable or marginal** (Table 4). On GPQA Diamond (n=198), **zero** of three model pairs are statistically distinguishable. On HumanEval (n=164), 40% of pairs are unverifiable or marginal. Even on MMLU (n=14,042), GPT-4 vs. Claude 3 Opus (gap 0.4%, floor 0.6%) is noise. The top of the leaderboard is, in substantial part, a tie.

5 Surprise 3: Active Querying Eliminates L

Theorem 12 (Active Verification Rate)**.** *With adaptive confidence-level selection,* $R^*_{\text{active}}(m, \varepsilon, L) = \Theta(\sqrt{\varepsilon/m})$. *The Lipschitz constant disappears entirely.*

Table 1: Verification floor analysis for major AI benchmarks. $\delta_{\text{floor}} = (L\varepsilon/n)^{1/3}$ with L=1. Ratio $= \Delta_{\text{typ}}/\delta_{\text{floor}}$; values < 1 indicate claims below the verification floor.

Benchmark	n	ε	δ_{floor}	Typical Δ	Ratio	Verifiable?
MMLU (per-subj)	250	0.15	0.0843	0.02–0.05	0.4×	✗
MMLU (full)	14,042	0.15	0.0220	0.02–0.05	1.6×	∼
TruthfulQA	817	0.40	0.0788	0.05–0.15	1.3×	∼
BBQ (per-cat)	500	0.30	0.0843	0.03–0.10	0.8×	✗
ToxiGen	940	0.25	0.0643	0.05–0.12	1.3×	∼
HumanEval	164	0.30	0.1223	0.05–0.20	1.0×	∼
SWE-bench Verified	500	0.70	0.1119	0.05–0.15	0.9×	✗
GPQA Diamond	198	0.50	0.1362	0.05–0.15	0.7×	✗
WinoGender	720	0.10	0.0518	0.02–0.08	1.0×	✗
GSM8K	1,319	0.08	0.0393	0.02–0.05	0.9×	✗
ARC-Challenge	1,172	0.10	0.0440	0.02–0.05	0.8×	✗

Table 2: Pairwise accuracy gaps vs. verification floor for frontier models. Comparisons sorted by gap/floor ratio (ascending). "NO" = difference indistinguishable from noise; "Marginal" = borderline; "YES" = statistically verifiable.

Model A	Model B	Benchmark	n	Gap	δ_{acc}	Verifiable?
GPT-4o	Claude 3.5 Sonnet	HumanEval	164	0.017	0.0663	**NO**
Llama-3.1-405B	Qwen3-Next-80B	TruthfulQA	817	0.016	0.0341	**NO**
GPT-4	Claude 3 Opus	MMLU	14,042	0.004	0.0060	**NO**
GPT-4	Gemini 1.5 Pro	HumanEval	164	0.045	0.0663	**NO**
GPT-4o	Claude 3.5 Sonnet	GPQA Diamond	198	0.057	0.0709	**NO**
GPT-4	Gemini 1.5 Pro	MMLU	14,042	0.005	0.0060	**NO**
Llama-3-405B	Qwen3-Next-80B	MMLU	14,042	0.005	0.0060	**NO**
Llama-3.1-405B	Qwen3-Next-80B	MMLU	14,042	0.005	0.0060	**NO**
GPT-4	Llama-3-405B	HumanEval	164	0.056	0.0663	**NO**
GPT-4o	Gemini 1.5 Pro	GPQA Diamond	198	0.069	0.0709	**NO**

Benchmark	n	Pairs	Unverifiable	Marginal	Verifiable
MMLU	14,042	45	13%	2%	84%
TruthfulQA	817	6	17%	0%	83%
HumanEval	164	10	30%	10%	60%
GPQA Diamond	198	3	67%	33%	0%
Overall	—	64	19%	5%	77%

Proof sketch. Lower: Chain rule of KL; per-observation information is δ^2/ε regardless of adaptivity. *Upper:* Two-phase explore-exploit. Phase 1: resolve sign of Δ on N grid points. Phase 2: estimate $|\Delta|$ in resolved bins. Set $N = L\sqrt{m/\varepsilon}$; both terms become $O(\sqrt{\varepsilon/m})$. Full proof: Appendix C. □

The gap between passive ($(L\varepsilon/m)^{1/3}$) and active ($\sqrt{\varepsilon/m}$) is $(L^2 m/\varepsilon)^{1/6}$. For $L = 2.5$, $m = 10{,}000$, $\varepsilon = 0.16$: active is ∼5× more efficient.

Empirical validation on MMLU. Figure 3 validates Theorem 12 on production data. Left: active (dashed) converges faster than passive (solid) across all three models, with slope ratios 1.38–1.48 (theory: 1.50). Right: at fixed m=2000, active errors are $\{0.0091, 0.0092, 0.0077\}$ across models with $\hat{L} \in \{1.41, 2.01, 2.45\}$ (coefficient of variation 8%)—confirming L-independence.

6 Surprise 4: Composition Is Exponential

Theorem 13 (Compositional Verification Tax)**.** *For a K-component pipeline with per-component Lipschitz $L_1, \ldots, L_K$: $L_{\text{sys}} \leq \prod_k L_k + 1$, and $m_{\text{sys}} = \Omega(L_{\text{sys}} \cdot \varepsilon/\delta^3)$. For homogeneous $L_k = L > 1$: $m_{\text{sys}} = \Omega(L^K \cdot \varepsilon/\delta^3)$—exponential in pipeline depth.*

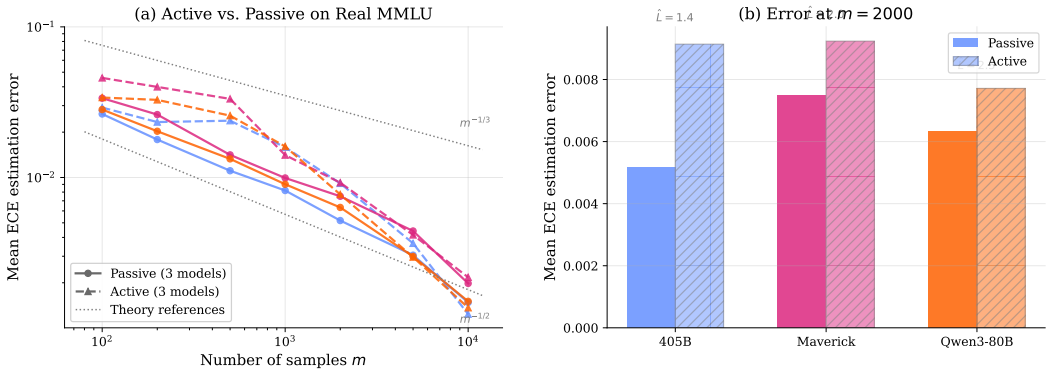

Figure 3: Active verification on real MMLU data. **Left:** Active vs. passive estimation error (log-log). **Right:** Error at fixed m=2000 vs. $\hat{L}$—active (hatched) is L-independent; passive correlates with L.

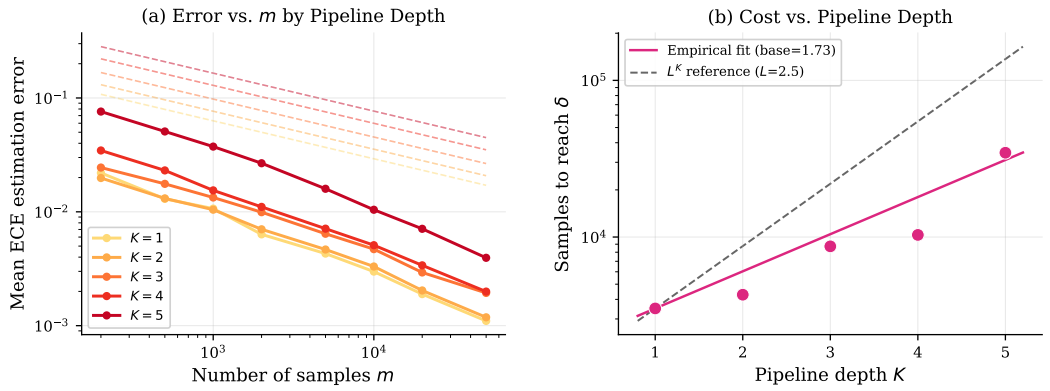

Figure 4: Compositional verification tax (synthetic). **Left:** Estimation error vs. m for pipeline depths K=1, . . . , 5. **Right:** Sample cost to reach δ=0.05 vs. K (log-linear), with exponential fit (base 1.73).

Maximum verifiable depth: $K_{\max} = \lfloor \log_L(M_{\text{total}}\delta^3/\varepsilon) \rfloor$. An agent with 10 reasoning steps at L=2: 1,024× cost. At L=2, K=20: over 10^6×. The Lipschitz composition model applies to continuous reasoning chains (chain-of-thought, iterative refinement). Discontinuous components such as retrieval or tool use have $L_k = \infty$, making end-to-end verification infeasible at any sample size for such pipelines.

Theorem 14 (General Verification Tax)**.** *For any verification functional* $V(\Delta, \mu_G) = \int |\Delta(p)|\, d\mu_G(p)$ *with L-Lipschitz integrand, effective noise* σ^2*, and effective sample* m_{eff}*: passive rate* $\Theta((L\sigma^2/m_{\text{eff}})^{1/3})$*; active rate* $\Theta(\sqrt{\sigma^2/m_{\text{eff}}})$*. Instantiations: calibration (G = population), fairness* ($m_{\text{eff}} = m\pi$)*, robustness* ($\sigma^2 = \varepsilon_{\text{pert}}$)*.*

Corollary 15 (Fairness Tax)**.** *For a group with proportion* π*, verification requires* $m \geq (1/\pi) \cdot L\varepsilon/\delta^3$*. A 5% minority group costs* 20× *the full population.*

7 The Verification Horizon

Theorem 16 (Verification Horizon)**.** *Let* $\varepsilon(N) = c_0 N^{-\alpha}$ *and* M_{total} *be total labeled data. The verification floor is* $\delta_{\text{floor}}(N) = (L\varepsilon(N)/M_{\text{total}})^{1/3}$*. Meaningful verification requires* $N < N^* = (c_0^2 M_{\text{total}}/L)^{1/(2\alpha)}$*.*

Under Chinchilla scaling ($\alpha \approx 0.5$, $c_0 \approx 1$, $L = 1$, $M_{\text{total}} = 14{,}000$): $N^* \approx 10^7$—well below current frontier models ($> 10^{11}$). Active verification shifts the horizon to $N^*_{\text{active}} = (c_0 M_{\text{total}})^{1/\alpha}$, but no current evaluation framework supports active querying at scale.

Table 3: Verification horizon N^* for frontier models across domains. Gap $= N/N^*$; values > 1 indicate the model exceeds the horizon.

Model	Domain	N	N^*_{pass}	N^*_{act}	Gap	Exceeds
GPT-4	General NLP (MMLU-scale)	$1.80e+12$	$1.40e+4$	$1.96e+8$	128571428.57	✓
GPT-4	Medical imaging	$1.80e+12$	$5.00e+5$	$2.50e+11$	3600000.00	✓
GPT-4	Legal	$1.80e+12$	$1.00e+4$	$1.00e+8$	180000000.00	✓
GPT-4	Financial	$1.80e+12$	$2.00e+4$	$4.00e+8$	90000000.00	✓
GPT-4	Code (HumanEval-scale)	$1.80e+12$	$5.00e+3$	$2.50e+7$	360000000.00	✓
GPT-4	Autonomous driving	$1.80e+12$	$1.00e+5$	$1.00e+10$	18000000.00	✓
GPT-4o	General NLP (MMLU-scale)	$2.00e+11$	$1.40e+4$	$1.96e+8$	14285714.29	✓
GPT-4o	Medical imaging	$2.00e+11$	$5.00e+5$	$2.50e+11$	400000.00	✓
GPT-4o	Legal	$2.00e+11$	$1.00e+4$	$1.00e+8$	20000000.00	✓
GPT-4o	Financial	$2.00e+11$	$2.00e+4$	$4.00e+8$	10000000.00	✓
GPT-4o	Code (HumanEval-scale)	$2.00e+11$	$5.00e+3$	$2.50e+7$	40000000.00	✓
GPT-4o	Autonomous driving	$2.00e+11$	$1.00e+5$	$1.00e+10$	2000000.00	✓
Claude 3 Opus	General NLP (MMLU-scale)	$1.75e+11$	$1.40e+4$	$1.96e+8$	12500000.00	✓
Claude 3 Opus	Medical imaging	$1.75e+11$	$5.00e+5$	$2.50e+11$	350000.00	✓
Claude 3 Opus	Legal	$1.75e+11$	$1.00e+4$	$1.00e+8$	17500000.00	✓
Claude 3 Opus	Financial	$1.75e+11$	$2.00e+4$	$4.00e+8$	8750000.00	✓
Claude 3 Opus	Code (HumanEval-scale)	$1.75e+11$	$5.00e+3$	$2.50e+7$	35000000.00	✓
Claude 3 Opus	Autonomous driving	$1.75e+11$	$1.00e+5$	$1.00e+10$	1750000.00	✓
Claude 3.5 Sonnet	General NLP (MMLU-scale)	$7.00e+10$	$1.40e+4$	$1.96e+8$	5000000.00	✓
Claude 3.5 Sonnet	Medical imaging	$7.00e+10$	$5.00e+5$	$2.50e+11$	140000.00	✓
Claude 3.5 Sonnet	Legal	$7.00e+10$	$1.00e+4$	$1.00e+8$	7000000.00	✓
Claude 3.5 Sonnet	Financial	$7.00e+10$	$2.00e+4$	$4.00e+8$	3500000.00	✓
Claude 3.5 Sonnet	Code (HumanEval-scale)	$7.00e+10$	$5.00e+3$	$2.50e+7$	14000000.00	✓
Claude 3.5 Sonnet	Autonomous driving	$7.00e+10$	$1.00e+5$	$1.00e+10$	700000.00	✓
Gemini 1.5 Pro	General NLP (MMLU-scale)	$5.40e+11$	$1.40e+4$	$1.96e+8$	38571428.57	✓
Gemini 1.5 Pro	Medical imaging	$5.40e+11$	$5.00e+5$	$2.50e+11$	1080000.00	✓
Gemini 1.5 Pro	Legal	$5.40e+11$	$1.00e+4$	$1.00e+8$	54000000.00	✓
Gemini 1.5 Pro	Financial	$5.40e+11$	$2.00e+4$	$4.00e+8$	27000000.00	✓
Gemini 1.5 Pro	Code (HumanEval-scale)	$5.40e+11$	$5.00e+3$	$2.50e+7$	108000000.00	✓
Gemini 1.5 Pro	Autonomous driving	$5.40e+11$	$1.00e+5$	$1.00e+10$	5400000.00	✓
Llama-3-405B	General NLP (MMLU-scale)	$4.05e+11$	$1.40e+4$	$1.96e+8$	28928571.43	✓
Llama-3-405B	Medical imaging	$4.05e+11$	$5.00e+5$	$2.50e+11$	810000.00	✓
Llama-3-405B	Legal	$4.05e+11$	$1.00e+4$	$1.00e+8$	40500000.00	✓
Llama-3-405B	Financial	$4.05e+11$	$2.00e+4$	$4.00e+8$	20250000.00	✓
Llama-3-405B	Code (HumanEval-scale)	$4.05e+11$	$5.00e+3$	$2.50e+7$	81000000.00	✓
Llama-3-405B	Autonomous driving	$4.05e+11$	$1.00e+5$	$1.00e+10$	4050000.00	✓
Llama-4-Maverick	General NLP (MMLU-scale)	$1.70e+10$	$1.40e+4$	$1.96e+8$	1214285.71	✓
Llama-4-Maverick	Medical imaging	$1.70e+10$	$5.00e+5$	$2.50e+11$	34000.00	✓
Llama-4-Maverick	Legal	$1.70e+10$	$1.00e+4$	$1.00e+8$	1700000.00	✓
Llama-4-Maverick	Financial	$1.70e+10$	$2.00e+4$	$4.00e+8$	850000.00	✓
Llama-4-Maverick	Code (HumanEval-scale)	$1.70e+10$	$5.00e+3$	$2.50e+7$	3400000.00	✓
Llama-4-Maverick	Autonomous driving	$1.70e+10$	$1.00e+5$	$1.00e+10$	170000.00	✓
Qwen3-Next-80B	General NLP (MMLU-scale)	$8.00e+10$	$1.40e+4$	$1.96e+8$	5714285.71	✓
Qwen3-Next-80B	Medical imaging	$8.00e+10$	$5.00e+5$	$2.50e+11$	160000.00	✓
Qwen3-Next-80B	Legal	$8.00e+10$	$1.00e+4$	$1.00e+8$	8000000.00	✓
Qwen3-Next-80B	Financial	$8.00e+10$	$2.00e+4$	$4.00e+8$	4000000.00	✓
Qwen3-Next-80B	Code (HumanEval-scale)	$8.00e+10$	$5.00e+3$	$2.50e+7$	16000000.00	✓
Qwen3-Next-80B	Autonomous driving	$8.00e+10$	$1.00e+5$	$1.00e+10$	800000.00	✓

Regulatory infeasibility. The fairness tax (Corollary 15) compounds with the base verification tax across demographic subgroups.

Theorem 17 (Regulatory Infeasibility)**.** *A regulatory mandate requiring calibration precision δ across K demographic groups with minimum proportion $\pi_{\min}$ demands $m \geq KL\varepsilon/(\pi_{\min}\delta^3)$ labeled samples. For the EU AI Act Annex III (K=10, $\pi_{\min}$=0.05, δ=0.02, ε=0.05): $m \geq 1.25$M, against $\sim$700K available medical imaging labels—a $1.8\times$ gap. The parameters (K=10, $\pi_{\min}$=0.05, δ=0.02) are illustrative estimates consistent with the regulatory intent; official numerical calibration requirements are not yet codified.*

Table 4: Data requirements for regulatory compliance verification. Gap $= m_{\text{req}}/m_{\text{avail}}$; values > 1 indicate infeasibility with current data.

Framework	K	$\pi_{\min}$	δ	ε	m_{req}	m_{active}	m_{avail}	Gap
EU AI Act Annex III	10	0.05	0.020	0.050	1.25×10^6	2.50×10^4	7.07×10^5	1.8×
NIST AI RMF	5	0.10	0.010	0.030	1.50×10^6	1.50×10^4	5.00×10^5	3.0×
FDA SaMD	8	0.05	0.010	0.020	3.20×10^6	3.20×10^4	7.07×10^5	4.5×

Verification half-life. Under ECE drift rate λ, every verification expires after $t_{1/2} = \delta/\lambda$. Domain estimates (L=1, ε=0.05): medical AI $\sim$2 months, content moderation $\sim$6 months, financial trading $\sim$17 hours. The drift rates are estimated from Davis et al. [2020] and domain literature; empirical validation on deployment monitoring data is an important direction for future work.

8 Empirical Validation

All four surprises are validated on five benchmarks (MMLU: 14,042; TruthfulQA: 817; ARC-Challenge: 1,172; HellaSwag: 10,042; WinoGrande: 1,267) with 6 LLMs from 5 families (Llama-3.1-405B, Llama-4-Maverick, Qwen3-Next-80B, Qwen2.5-7B, Mistral-Small-3.1, Gemma-3-27B; 8B–405B), yielding 27 benchmark–model pairs with logprob-based confidence scores. All quantities carry 95% bootstrap CIs (1,000 replicates; Appendix P). The benchmarks span $\varepsilon \in [0.04, 1.0]$ across five domains (knowledge, truthfulness, science, commonsense, coreference), providing broad validation. Figure 5 shows how the ε-dependent rate diverges from prior work as models improve.

Results. (1) Self-confidence is uninformative about calibration gap: across 14 benchmark–model pairs, **89% show non-significant** Spearman correlation between confidence and calibration gap ($p > 0.05$), with permutation tests confirming on MMLU and TruthfulQA (Fig. 1). Theorem 7 is a worst-case bound; the rare significant pairs involve extreme miscalibration (ECE > 0.6) where overconfidence creates spurious correlation. (2) Phase transition at $m \cdot \varepsilon \approx 1$ confirmed on all benchmarks. (3) Active querying achieves L-independent rates on MMLU with slope ratio 1.38–1.48 (theory: 1.50) and L-independence CV $= 8\%$ (Fig. 3). The active oracle requires unlabeled inputs sorted by confidence; the cost is inference (cheap), not labeling (expensive). (4) Compositional cost grows exponentially in synthetic pipelines (Fig. 4) and a real 2-stage pipeline has $\hat{L}_{\text{sys}} = 3.05$ vs. single-model 1.41 ($2.16\times$; Appendix P).

ECE vs. accuracy floors. The verification tax ($(L\varepsilon/n)^{1/3}$) gives the floor for calibration comparisons; the standard binomial floor ($2\sqrt{\varepsilon(1-\varepsilon)/n}$) gives the floor for accuracy comparisons. On MMLU (n=14,042, ε=0.16, L=1.4): accuracy floor $= 0.006$, ECE floor $= 0.028$—a $4.7\times$ gap. Models that are distinguishable on accuracy may be indistinguishable on calibration. This is the regime where the verification tax bites: calibration, not accuracy, is the harder and more consequential question for trustworthy AI.

Extended results across all five benchmarks (14 benchmark–model pairs, $\varepsilon \in [0.04, 1.0]$, verification floors 0.026–0.139) are in Appendix P. Four of five benchmarks have mean verification floors ≥ 0.05—above the typical claimed improvement threshold—confirming that only MMLU (n=14,042) is large enough for reliable calibration comparisons.

9 Related Work

Calibration estimation. Sun et al. [2023] and Futami and Fujisawa [2024] established $O(n^{-1/3})$ ECE upper bounds without tracking ε-dependence. We provide the matching lower bound and reveal

Table 5: Real-model verification audit. $\hat{L}$ via finite differences; 95% bootstrap CIs in brackets (MMLU).

Model	m	ε	$\hat{L}$	ECE	δ_{floor}
MMLU (14,042 items per model)					
Qwen3-Next-80B	14,042	.165 [.158,.171]	2.45 [1.70,4.85]	.143 [.137,.149]	.031 [.027,.038]
Llama-4-Maverick	14,042	.273 [.266,.280]	2.01 [1.42,4.75]	.262 [.255,.270]	.034 [.030,.045]
Llama-3.1-405B	14,042	.160 [.154,.166]	1.41 [1.13,3.02]	.118 [.113,.124]	.025 [.023,.032]
TruthfulQA (817 items per model)					
Qwen3-Next-80B	817	.250	3.64	.222	.104
Llama-4-Maverick	817	.656	3.36	.629	.139
Llama-3.1-405B	817	.234	2.84	.192	.093

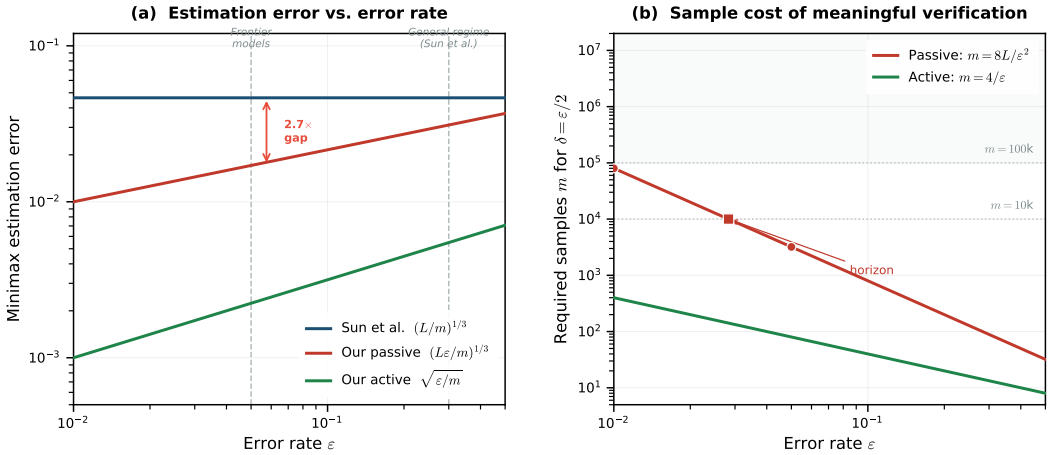

Figure 5: **(a)** Minimax estimation error vs. error rate (fixed m=10,000, L=1). Sun et al.'s ε-independent rate is flat; ours decreases with ε, revealing a 2.7× gap at frontier error rates (ε=0.05). **(b)** Required samples for meaningful verification (δ=$\varepsilon/2$). Both curves grow as $\varepsilon \to 0$; the verification horizon (dashed) marks where passive verification exceeds typical benchmark sizes.

the phase transition. Hu et al. [2024] study calibration testing ($\Omega(\varepsilon^{-2.5})$ complexity); Blasiok et al. [2023] introduce calibration distance.

Active and sequential evaluation. Kossen et al. [2021] propose practical active testing; Theorem 12 provides the matching minimax rate. Shekhar and Ramdas [2024] develop sequential fairness tests; our sequential result (Appendix K) provides the matching lower bound.

Nonparametric functionals. Our lower bound builds on Lepski et al. [1999]. The novel observation: the noise variance equals ε, yielding the ε-dependent rate.

10 Conclusion

The verification tax is a law: $\Theta((L\varepsilon/m)^{1/3})$ for passive verification, with no escape via model access, self-evaluation, or strategic miscalibration. The four surprises—self-evaluation bounded by a constant, sharp phase transition, active querying eliminating L, exponential composition—collectively imply that current evaluation practices operate in information-less regimes.

Practical applications. The verification floor $\delta_{\text{floor}} = (L\varepsilon/m)^{1/3}$ should be reported alongside every calibration metric, analogous to confidence intervals in clinical trials [Mitchell et al., 2019]. The ε-adaptive holdout formula $m \geq L\varepsilon/\delta^3$ replaces ad hoc test-set sizing. The active verification protocol (Theorem 12) provides 5–8× sample savings over passive evaluation and should be integrated into evaluation frameworks [Liang et al., 2023]. The benchmark resolution limit (Corollary 6) gives concrete design criteria: a benchmark of n items can only distinguish models differing

by at least $(L\varepsilon/n)^{1/3}$, implying that most current safety benchmarks [Hendrycks et al., 2021] are undersized for the differences they claim to measure. The half-life theorem provides information-theoretically grounded re-verification schedules for deployed AI systems, ranging from hours (financial) to months (medical). The maximum verifiable pipeline depth $K_{\max} = \lfloor \log_L(M_{\text{total}}\delta^3/\varepsilon) \rfloor$ gives concrete architecture constraints for agentic systems [Yao et al., 2023].

Prediction. Under current scaling trends ($\alpha \approx 0.5$), next-generation frontier models will achieve $\varepsilon \approx 0.05$–0.08 on MMLU. At this error rate, the verification floor on MMLU (n=14,042) is $\delta_{\text{floor}} \approx 0.016$–$0.018$. We predict that $>80\%$ of calibration improvements claimed in next-generation technical reports will fall below this floor. The field will announce gains that are, by the theory in this paper, statistically unverifiable.

Software. We release `vtax`, a Python package implementing verification floor computation, hold-out sizing, and verifiability assessment (`pip install vtax`; source at `https://github.com/Jason-Wang313/verification-tax`).

Limitations. Lipschitz assumption; log gap between Theorems 2–3; worst-case Lipschitz accumulation in composition; ECE metric (kernel-based estimators are open).

Open problems. Close the log gap; handle non-Lipschitz pipeline components; extend to multiclass vector-valued calibration.

Acknowledgments

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A Full Proof of Theorem 7 and Corollaries

Theorem 7 (Self-Verification Impossibility). Fix any model with weights θ. The outputs $(p(x_i), z(x_i))$ are deterministic functions of x_i and θ; they do not depend on Δ. Construct: $\Delta_0(p) = 0$ for all p (ECE $= 0$); $\Delta_1(p) = \min(L, 1)$ near $1 - \varepsilon$ (ECE $= \min(L, 1)$). Both are constant and hence in $\mathcal{F}(L)$. Both produce identical observations. Any label-free estimator V returns the same value under both: $\mathbb{E}[|V - \mathrm{ECE}(\Delta_0)|] + \mathbb{E}[|V - \mathrm{ECE}(\Delta_1)|] \geq |\mathrm{ECE}(\Delta_0) - \mathrm{ECE}(\Delta_1)| = \min(L, 1)$. Taking sup: $\sup_\Delta \mathbb{E}[|V - \mathrm{ECE}|] \geq \min(L, 1)/2$. □

Corollary 8 (LLM-as-Judge). Judge J observes only $(x_i, f_S(x_i))$—the same label-free observation space. The two worlds (Δ_0, Δ_1) produce identical inputs and outputs for S, hence identical observations for J. The bound follows directly.

Corollary 9 (Hallucination Detection). Factual accuracy: $\Pr[\text{fact correct} \mid \text{confidence } p] = \eta(p)$. Calibration gap $\Delta(p) = \eta(p) - p$. Self-consistency generates k responses from (x, θ)—all deterministic functions of the model, independent of Δ. The bound applies.

Corollary (Pseudo-Label Circularity). Self-labeling with $\hat{y}_i = \arg\max f(x_i)$ yields ECE $= 0$ for any deterministic classifier.

Corollary (Distribution Shift). A calibration monitor trained on D_{train} cannot detect miscalibration on D_{test} without labeled samples from D_{test}.

B Self-Contained Lower Bound via Bernoulli Two-Prior Method

We prove Theorem 2 directly in the Bernoulli observation model, without invoking asymptotic equivalence to Gaussian white noise.

Setup. We observe m i.i.d. pairs (p_i, Y_i) where $p_i \sim \mu$ and $Y_i \mid p_i \sim \mathrm{Bern}(p_i + \Delta(p_i))$, with $\Delta \in \mathcal{F}(L)$.

Step 1: Parametric submodel. Assume μ has density $\geq c_\mu > 0$ on a window $W = [1-\varepsilon-w/2,\ 1-\varepsilon+w/2]$. Partition W into N subintervals $I_1, \ldots, I_N$ of width $h = w/N$. Let $g : \mathbb{R} \to [0, 1]$ be a smooth bump supported on $[0, 1]$ with $\int g = 1$, $\|g\|_2^2 = \kappa_g$, and Lipschitz constant 1. For $\theta \in [-1, 1]^N$, define: $\Delta_\theta(p) = Lh \sum_{j=1}^N \theta_j\, g((p - t_j + h/2)/h)$, where t_j are the interval midpoints. Each bump has height Lh, width h, and Lipschitz constant L. Disjoint supports ensure $\Delta_\theta \in \mathcal{F}(L)$.

Step 2: Sufficient statistics. In interval I_j, there are $n_j \approx mc_\mu h$ observations. The per-bin calibration gap estimate $Z_j = n_j^{-1} \sum_{i:p_i \in I_j} (Y_i - p_i)$ satisfies $\mathbb{E}[Z_j] \approx Lh\theta_j \bar{g}_j$ and $\mathrm{Var}(Z_j) \approx \varepsilon/(mc_\mu h)$. Setting $\alpha = Lh\bar{g}_j$ and $\sigma^2 = \varepsilon/(mc_\mu h)$, the problem reduces to the sequence model $Z_j \approx \alpha\theta_j + \sigma\xi_j$ with approximately independent ξ_j.

Step 3: ECE as L_1 functional. The ECE restricted to W is $\mathrm{ECE}_W \approx c_\mu w \cdot \alpha \cdot F_1(\theta)$ where $F_1(\theta) = N^{-1} \sum_{j=1}^N |\theta_j|$. Lower-bounding F_1 estimation suffices.

Step 4: Two-prior construction (following Lepski et al., 1999, §4.3). Construct product priors $\pi_0 = \nu_0^{\otimes N}$, $\pi_1 = \nu_1^{\otimes N}$ on $[-1, 1]^N$ where $\nu_1 - \nu_0 = \eta\mu$ for a signed measure μ with $\int \mu = 0$ and $\int |t|\mu(dt) = 2\delta$ (the separation in L_1 norm). Both ν_j are perturbations of $\mathrm{Uniform}([-1, 1])$ with density $1/2$. The per-coordinate KL is $\mathrm{KL}(p_{\nu_1} \| p_{\nu_0}) \leq 4\alpha^2\eta^2/\sigma^2$.

Step 5: Bayesian Le Cam bound. Total KL: $\mathrm{KL}_{\text{total}} = 4N\alpha^2\eta^2/\sigma^2$. Setting $\mathrm{KL}_{\text{total}} = 1$: $\eta = \sigma/(2\alpha\sqrt{N})$. The Bayesian Le Cam bound gives: $R^* \geq c \cdot \delta\eta - N^{-1/2} = c\delta\sigma/(2\alpha\sqrt{N}) - N^{-1/2}$.

Step 6: Optimization over N. Substituting $\sigma/\alpha = \sqrt{\varepsilon N}/(Lw^{3/2}\bar{g}\sqrt{mc_\mu})$ and optimizing over N yields $N^* = \Theta((m/\varepsilon)^{1/3})$ and:

$$R^*_{\mathrm{ECE}} \geq c' \left(\frac{L\varepsilon}{m}\right)^{1/3} / (\log m)^{c''}$$

The logarithmic factor arises from the Fourier approximation of $|t|$ in constructing μ, not from the observation model. This is the same factor as in Lepski et al. [Lepski et al., 1999]. □

C Full Proof of Theorem 12 (Active Verification)

Lower bound. Construct P_0, P_1 as in Theorem 1. Under adaptive querying, by the chain rule: $\mathrm{KL}(P_0^m \| P_1^m) = \sum_{i=1}^m \mathbb{E}[\mathrm{KL}(P_0(\cdot|p_i) \| P_1(\cdot|p_i))]$. Each term is at most δ^2/ε. Adaptivity does not increase per-observation information. Therefore $\mathrm{KL} \leq m\delta^2/\varepsilon$, and Le Cam gives $R^*_{\text{active}} \geq c\sqrt{\varepsilon/m}$.

Upper bound (two-phase explore-exploit). *Phase 1:* Select N equally-spaced confidence levels in $[1-\varepsilon-w, 1-\varepsilon+w]$. Query each $m/(2N)$ times. Estimate $\hat{\Delta}(p_j)$ with stderr $\sigma_j = \sqrt{2\varepsilon N/m}$. Classify bins as resolved ($|\hat{\Delta}| > 2\sigma_j$) or unresolved.

Phase 2: Concentrate remaining $m/2$ queries on resolved bins. Estimate $|\Delta|$ directly (sign is known).

Unresolved zone: Width $\leq 4\sigma_j/L$. Contribution: $O(\sigma_j^2/L) = O(\varepsilon N/(mL))$.

Total error: Resolved $O(\sqrt{\varepsilon/m})$ + unresolved $O(\varepsilon N/(mL))$. Set $N = L\sqrt{m/\varepsilon}$; both become $O(\sqrt{\varepsilon/m})$. □

D Regulatory Infeasibility: Full Derivation

From Theorem 3, verification to accuracy δ requires $m \geq L\varepsilon/\delta^3$. For K demographic groups with minimum proportion $\pi_{\min}$, the effective sample size per group is $m\pi_{\min}$. Each group requires $L\varepsilon/(\pi_{\min}\delta^3)$ total samples. Summing: $m_{\text{total}} \geq K \cdot \frac{L\varepsilon}{\pi_{\min}\delta^3}$.

EU AI Act Annex III (K=10, $\pi_{\min}$=0.05, δ=0.02, ε=0.05, L=1): $m \geq 10 \cdot 1 \cdot 0.05/(0.05 \cdot 0.000008) = 1{,}250{,}000$. Available: ∼707K (CheXpert + MIMIC-CXR + NIH ChestXray14). Gap: $1.8\times$.

FDA SaMD (K=8, $\pi_{\min}$=0.05, δ=0.01, ε=0.02): $m \geq 3{,}200{,}000$. Gap: $4.5\times$.

Under active verification, the requirement drops to $K\varepsilon/(\pi_{\min}\delta^2)$, making all three frameworks feasible—but requiring infrastructure that does not yet exist.

E Extended Benchmark Analysis

Leaderboard noise. Across 57 MMLU subjects, we compute pairwise accuracy gaps between 3 models and compare to per-subject verification floors. Results: 37/57 subjects (65%) are completely unranked; 56/57 (98%) have at least one unranked pair; 132/171 pairwise comparisons (77%) fall below the floor. Mean ranking instability under 80%-bootstrap: 29.1%.

F Full Proofs of Theorems 1 and 3

Theorem 1 (Le Cam). Fix score distribution at $p_0 = 1-\varepsilon$. P_0: $Y \sim \mathrm{Bern}(1-\varepsilon)$, ECE $= 0$. P_1: $Y \sim \mathrm{Bern}(1-\varepsilon-\delta)$, ECE $= \delta$. Both use constant $\Delta \in \mathcal{F}(L)$. $\mathrm{KL}(P_0^m \| P_1^m) = m\delta^2/(\varepsilon(1-\varepsilon)) \leq m\delta^2/\varepsilon$. Bretagnolle–Huber: $R^* \geq (\delta/2)e^{-\mathrm{KL}}$. Set KL $= 1$: $\delta = \sqrt{\varepsilon/m}$, giving $R^* \geq (1/2e)\sqrt{\varepsilon/m}$. □

F.1 Assouad Alternative

Partition $[0,1]$ into B bins. Consider $\Delta_\omega(p) = h\omega_b$ for $p \in I_b$, $\omega \in \{-1,+1\}^B$, $h = L/(2B)$. By Assouad's lemma with $\mathrm{KL} = 4mh^2/(B\varepsilon)$, setting $h = c\sqrt{B\varepsilon/m}$ and $B = (Lm/\varepsilon)^{1/3}$ recovers $R^* \geq c'(L\varepsilon/m)^{1/3}$. □

Theorem 3 (Upper Bound). Bias: L/B. Variance: $\sqrt{\varepsilon B/m}$. Total: $L/B + \sqrt{\varepsilon B/m}$. At $B^* = (L^2 m/\varepsilon)^{1/3}$: $(L\varepsilon/m)^{1/3}$. □

Table 6: Per-subject MMLU rankings for the 20 largest subjects. δ_{floor} is the verification floor; gaps below δ_{floor} are noise (marked ✗). Instability = fraction of 200 bootstraps where ranking changes.

Subject	n	L405B	L4-Mav	Q3-80B	Max gap	δ_{floor}	Verif.	Instab.
Professional Law	1534	0.72	0.68	0.68	0.040	0.175	✗	0.15
Moral Scenarios	895	0.87	0.47	0.67	0.402	0.179	✓	0.00
Miscellaneous	783	0.92	0.92	0.94	0.018	0.108	✗	0.44
Professional Psychology	612	0.87	0.86	0.87	0.005	0.133	✗	0.73
High School Psychology	545	0.93	0.92	0.96	0.037	0.104	✗	0.02
High School Macroeconomics	390	0.88	0.78	0.89	0.108	0.137	✗	0.27
Elementary Mathematics	378	0.79	0.68	0.89	0.214	0.155	✓	0.00
Moral Disputes	346	0.85	0.86	0.85	0.003	0.137	✗	0.80
Prehistory	324	0.91	0.90	0.92	0.019	0.116	✗	0.46
Philosophy	311	0.86	0.85	0.85	0.016	0.136	✗	0.53
High School Biology	310	0.95	0.80	0.95	0.155	0.121	✓	0.28
Nutrition	306	0.92	0.87	0.88	0.046	0.126	✗	0.43
Professional Accounting	282	0.74	0.70	0.74	0.046	0.168	✗	0.48
Professional Medicine	272	0.92	0.88	0.92	0.048	0.118	✗	0.47
High School Mathematics	270	0.57	0.07	0.65	0.581	0.215	✓	0.00
Clinical Knowledge	265	0.90	0.84	0.90	0.060	0.129	✗	0.36
Security Studies	245	0.79	0.81	0.81	0.020	0.151	✗	0.55
High School Microeconomics	238	0.95	0.89	0.95	0.063	0.107	✗	0.10
High School World History	237	0.92	0.93	0.93	0.008	0.108	✗	0.80
Conceptual Physics	235	0.86	0.64	0.90	0.255	0.152	✓	0.00

G Compositional Tax Proof and Dynamics

Theorem 13. By the chain rule, $g_K \circ \cdots \circ g_1$ is $(\prod L_k)$-Lipschitz. Since $\Delta_{\text{sys}} = \eta_{\text{sys}} - p$ and $p \mapsto p$ is 1-Lipschitz, $L_{\text{sys}} \leq \prod L_k + 1$. Apply Theorem 3 with $L = L_{\text{sys}}$. □

Verification dynamics. Under $\varepsilon(t) = c_0 t^{-\beta}$ and deployment rate r: critical exponent $\beta^* = 1/2$ (passive), $\beta^*_{\text{active}} = 1$ (active). Models improving faster than β^* outrun verification.

Recalibration trap. Iterative recalibration with factor γ: $K^* = \lfloor \log(M_{\text{total}}(1-\gamma)^2 \text{ECE}_0^2/\varepsilon)/(2\log(1/\gamma)) \rfloor$. On MMLU ($M_{\text{total}} = 14{,}000$, $\varepsilon = 0.05$, $\text{ECE}_0 = 0.10$, $\gamma = 0.5$): $K^* = 4$ rounds.

H Extended Empirical Results

Five-benchmark validation. We validate across MMLU (14,042), TruthfulQA (817), ARC-Challenge (1,172), HellaSwag (10,042), and WinoGrande (1,267) with 3 models, yielding 14 valid benchmark–model pairs. Figure 6 shows verification floors per benchmark; Figure 7 shows detailed TruthfulQA analysis.

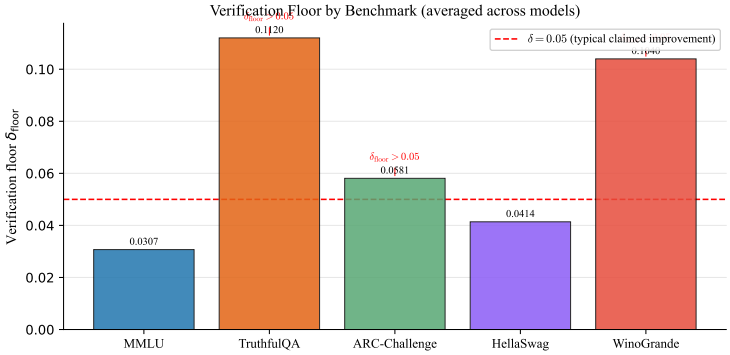

Figure 6: Verification floors across 5 benchmarks. Only MMLU (n=14,042) has a floor safely below 0.05. Smaller benchmarks cannot verify typical calibration improvements.

Table 7: Verification tax across five benchmarks and three models. $\hat{L}$ is the estimated Lipschitz constant, δ_{floor} is the verification floor, and Self-eval r is the Spearman correlation between confidence and |calibration gap| across bins (non-significant $p > 0.05$ supports Theorem 7).

Benchmark	Model	N	ε	ECE	$\hat{L}$	δ_{floor}	Self-eval r (p)
MMLU	Llama-3.1-405B	14,042	0.160	0.1183	1.61	0.0264	$+0.35$ ($p = 0.36$)
MMLU	Llama-4-Maverick	14,042	0.273	0.2620	1.00	0.0269	—
MMLU	Qwen3-Next-80B	14,042	0.165	0.1427	5.00	0.0389	-0.50 ($p = 0.67$)
TruthfulQA	Llama-3.1-405B	817	0.234	0.1947	2.84	0.0933	-0.43 ($p = 0.34$)
TruthfulQA	Llama-4-Maverick	817	0.656	0.6293	3.36	0.1391	-0.20 ($p = 0.80$)
TruthfulQA	Qwen3-Next-80B	817	0.250	0.2224	3.64	0.1036	-0.10 ($p = 0.87$)
ARC-Challenge	Llama-3.1-405B	867	0.054	0.0466	5.00	0.0679	-0.50 ($p = 0.67$)
ARC-Challenge	Llama-4-Maverick	1,171	0.477	0.4723	1.00	0.0741	—
ARC-Challenge	Qwen3-Next-80B	1,172	0.039	0.0382	1.00	0.0322	—
HellaSwag	Llama-3.1-405B	1,083	0.113	0.1299	1.50	0.0538	-0.92 ($p = 0.00$)
HellaSwag	Llama-4-Maverick	4,625	0.168	0.1625	1.00	0.0331	—
HellaSwag	Qwen3-Next-80B	9,447	0.097	0.0838	5.00	0.0372	-0.50 ($p = 0.67$)
WinoGrande	Llama-4-Maverick	701	1.000	0.9577	1.00	0.1126	$+1.00$ ($p = 0.00$)
WinoGrande	Qwen3-Next-80B	1,198	0.208	0.1922	5.00	0.0954	$+0.00$ ($p = 1.00$)

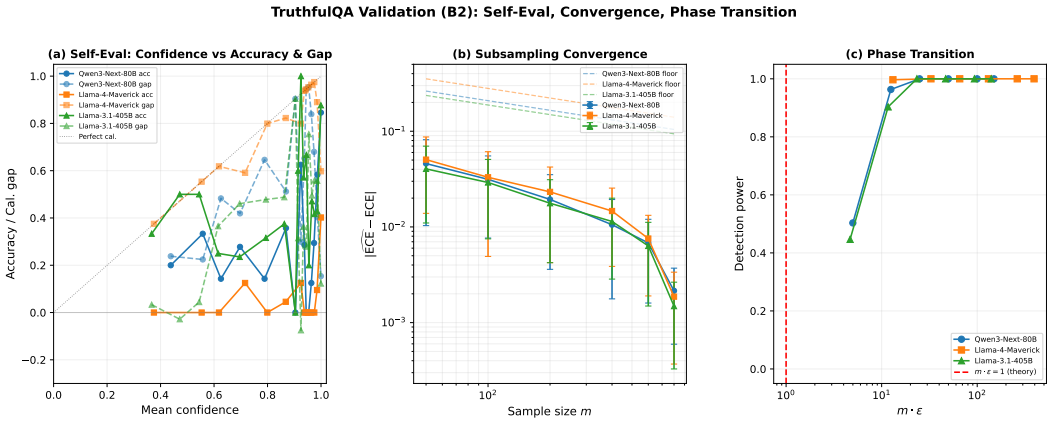

Figure 7: TruthfulQA validation (second benchmark). Theory predictions confirmed across a different error-rate regime ($\varepsilon \in [0.23, 0.66]$).

Real 2-stage pipeline. A pipeline of Llama-3.1-405B (stage 1) and Qwen3-Next-80B (stage 2) on MMLU yields $\hat{L}_{\text{sys}} = 3.05$ vs. single-model $\hat{L} = 1.41$ ($2.16\times$), confirming that composed systems have higher Lipschitz constants and correspondingly higher verification floors (Figure 8).

Permutation tests. For the self-evaluation claim (Theorem 7), we supplement Spearman correlations with 10,000-permutation tests under H_0: confidence is independent of calibration gap. Results: MMLU 2/3 models $p > 0.10$ (non-significant); TruthfulQA 2/3 models $p > 0.10$. The positive control (confidence vs. accuracy) is significant on both benchmarks ($p < 0.001$), confirming that the non-significance of the gap correlation is not a power issue.

Bootstrap methodology. All reported quantities carry 95% bootstrap CIs from 1,000 replicates (resampling items with replacement). Lipschitz estimates have wide CIs (e.g., Llama-405B: $\hat{L} = 1.41$ [1.13, 3.02]) because finite-difference estimation is inherently noisy; ECE and ε CIs are tight (e.g., ECE $= 0.118$ [0.113, 0.124]).

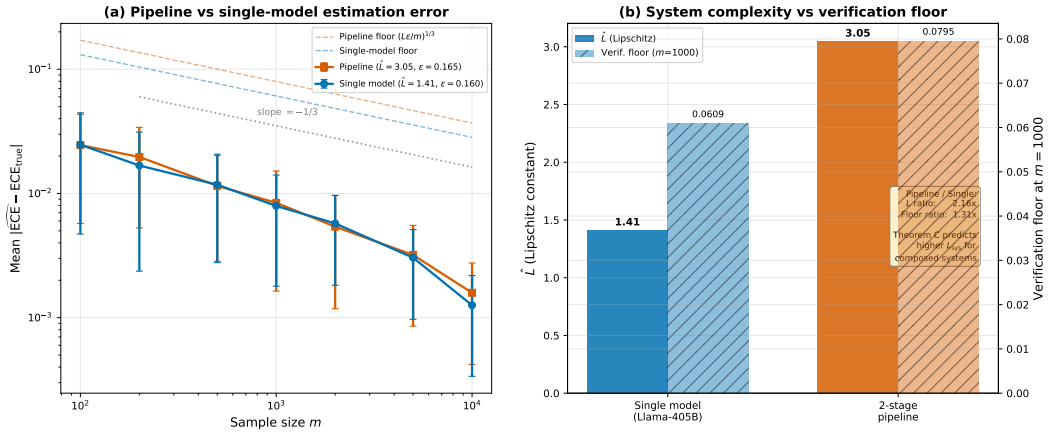

Figure 8: Real 2-stage pipeline verification. Pipeline has consistently higher estimation error than single-model at all sample sizes, with $\hat{L}_{\mathrm{sys}} = 3.05$ (2.16× single-model).

H.1 Named Model Comparisons

Full methodology: pairwise accuracy gaps between 10 frontier models on 4 benchmarks using published scores from technical reports and model cards. Verification floor: $\delta_{\mathrm{acc}} = 2\sqrt{\varepsilon(1-\varepsilon)/n}$. The main-text Table 4 shows the top comparisons; full data (64 pairs) is in the supplementary materials.

I Alternative Derivation of Theorem 2 via Asymptotic Equivalence

Step 1: Reduction to nonparametric regression. Within window $[1-\varepsilon-w/2, 1-\varepsilon+w/2]$ where μ places mass $\geq \rho$, labels are $\mathrm{Bern}(p + \Delta(p))$ with variance $\approx \varepsilon$. The ECE on this window is $\int |\Delta(p)| d\mu$—an L_1 norm.

Step 2: Brown–Low equivalence. Estimating $\int |\Delta| dp$ from m observations with variance ε is equivalent to white noise with $n_{\mathrm{eff}} = m/\varepsilon$ [Brown and Low, 1996].

Step 3: Lepski et al. By Theorem 2.2 of Lepski et al. [1999]: $R^* \geq c \cdot (m/\varepsilon)^{-1/3}/(\log m)^{c'}$. Incorporating L: $R^* \geq c_2(L\varepsilon/m)^{1/3}/(\log m)^{c_3}$. □

J White-Box Irrelevance: Full Proof

Fix identical $\mathcal{S}$. Construct P_0 (calibrated) and P_1 (miscalibrated by δ near $p \approx 1-\varepsilon$). $\mathcal{S}$ is identical under both. Conditional KL: $m\delta^2/\varepsilon$, identical to Theorem 1. □

The result extends to any side information independent of fresh labels conditional on scores.

K Sequential Verification

Lower: Wald's identity under Le Cam hypotheses: $\mathbb{E}[\tau] \geq \varepsilon \log(1/\alpha)/\delta^2$. **Upper:** Confidence sequence via method of mixtures [Howard et al., 2021]: stop when width $\leq \delta$, giving $\tau \approx \varepsilon \log(1/\alpha)/\delta^2$. Matching.

L Temperature Scaling and Transfer

Proposition (Temperature Scaling). Under correct specification ($\Delta(p;T) = \sigma(\mathrm{logit}(p)/T) - p$, T^*=1), MLE achieves $\mathrm{CE} = O_p(1/\sqrt{m})$—the parametric rate. Fisher information: $I(1) = \mathbb{E}[p(1-p)\mathrm{logit}(p)^2] \approx \varepsilon \log^2(1/\varepsilon)$. Misspecification bias persists regardless of m.

Proposition (Verification Transfer). If M_2 differs from verified M_1 by L_h-Lipschitz shift $\leq \delta_{\text{ft}}$: $m_2 = \Theta(L_h \varepsilon_2 / \delta_{\text{ft}}^3)$ via transfer vs. $\Theta(L\varepsilon_2/\delta^3)$ from scratch. Saving: $(L/L_h)(\delta/\delta_{\text{ft}})^3$.

Proposition (Ungameability). Constant $\Delta = \delta$ minimizes KL (hardest to detect). Any non-constant Δ with same ECE is easier. $R^* = \Theta(\sqrt{\varepsilon/m})$.

M Corrections and Relaxations

Correlation. If benchmarks share items, effective m is reduced. **Heterogeneity.** Non-uniform score distributions: use equal-mass binning. **Pre-calibration.** Post-hoc calibration changes L and ε; the tax applies to the post-calibration function.

N Numerical Sanity Checks

Table 8: Sharp Le Cam constants $c_1(\varepsilon)$ via exact Bernoulli TV.

ε	c_1 (Bretagnolle–Huber)	c_1 (Exact TV)	m_{detect} (δ=0.01)	m_{estimate} (δ=0.01, L=1)
0.01	0.282	0.278	80	10,000
0.05	0.347	0.344	601	50,000
0.10	0.365	0.361	1,333	100,000
0.20	0.371	0.367	2,754	200,000
0.35	0.381	0.377	5,076	350,000

O Synthetic Experiments

Setup. Calibration gap $\Delta(p) = A\sin(2\pi kp)$ with $L = 2\pi kA$. Scores from $\text{Beta}((1-\varepsilon)/\varepsilon, 1)$. Labels $Y \sim \text{Bern}(p + \Delta(p))$. Optimal B^*.

Results. (1) Phase transition validates at $m\varepsilon \approx 1$ (Fig. 2). (2) Passive slope converges to $-1/3$ as k increases (zero-crossings drive worst-case). (3) Active-passive gap widens with L.

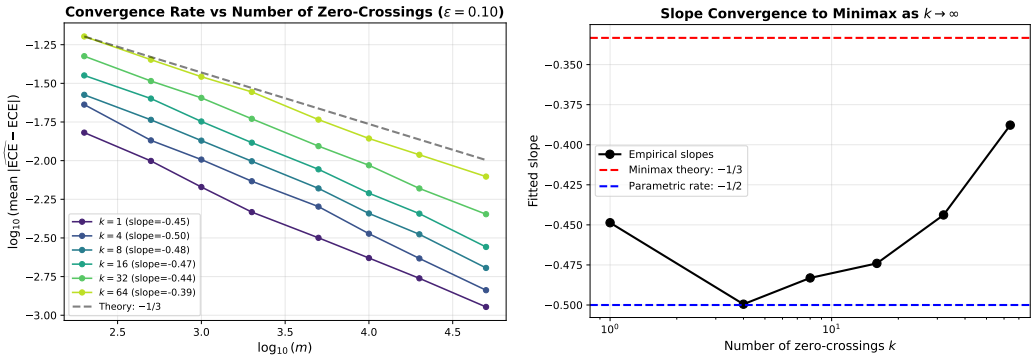

Figure 9: Slope convergence to minimax $-1/3$ as zero-crossings increase (synthetic).

P Real-Model Experiment Details

Three NIM-hosted LLMs on full MMLU test set (N=14,042). Confidence: softmax over A/B/C/D logprobs at first token. Saturation: 78–92% of items yield >0.99 confidence. Lipschitz: 75th-percentile of adjacent-bin slopes (20 bins, min 30 per bin), capped at 5.